

Ex.no:8	MINI PROJECT – FINANCIAL DATA ANALYTICS

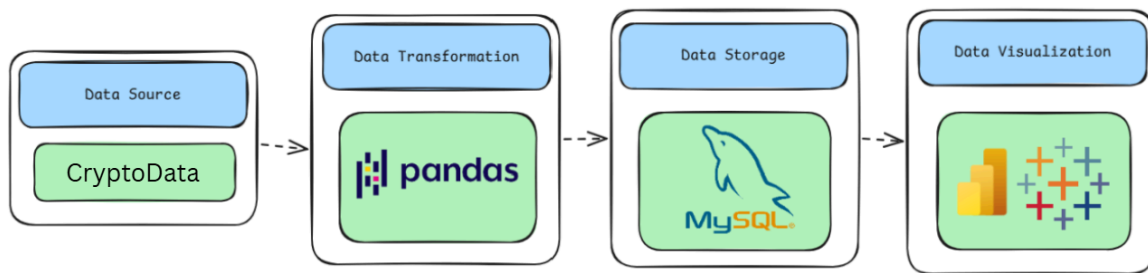
Aim:

To analyze cryptocurrency financial data using Python and Power BI by cleaning and transforming the dataset and creating interactive visualizations to study cryptocurrency prices, trading volume, market capitalization, returns, and volatility.

Algorithm:

1. Import the cryptocurrency dataset using Python and Pandas.
2. Inspect the dataset for missing values, duplicate records, and data types.
3. Remove duplicate records and convert the date and numerical columns into appropriate formats.
4. Handle missing numerical values using median values.
5. Calculate Daily Return using the opening and closing prices.
6. Calculate Price Range using the difference between high and low prices.
7. Calculate Volatility using the high, low, and opening prices.
8. Save the cleaned and transformed dataset as a CSV file.
9. Import the cleaned dataset into Power BI.
10. Create DAX measures for total records, total market capitalization, total trading volume, average closing price, average daily return, and average volatility.
11. Create KPI cards to display important financial indicators.
12. Create line charts to analyze cryptocurrency price trends over time.
13. Create bar charts to compare market capitalization, trading volume, and volatility among cryptocurrencies.
14. Create a scatter plot to analyze the relationship between trading volume and closing price.
15. Add slicers and filters for cryptocurrency and date to make the dashboard interactive.
16. Create a drill-through page to view detailed information for individual cryptocurrencies.
17. Analyze the dashboard visualizations and identify trends, performance differences, and risk patterns.

Architecture:



Program:

```
import pandas as pd
```

```
# Load dataset
```

```
df = pd.read_csv("C:/Users/91754/Desktop/dvt/consolidated_coin_data.csv")
```

```
# Create a copy for processing
```

```
crypto = df.copy()
```

```
# Remove duplicate records
```

```
crypto.drop_duplicates(inplace=True)
```

```
# Convert date to datetime
```

```
crypto["Date"] = pd.to_datetime(crypto["Date"], errors="coerce")
```

```
# Convert numerical columns
```

```
numeric_columns = [
```

```
    "Open",
```

```
    "High",
```

```
"Low",  
"Close",  
"Volume",  
"Market Cap"  
]
```

```
for col in numeric_columns:
```

```
    crypto[col] = (  
        crypto[col]  
        .astype(str)  
        .str.replace(",", "", regex=False)  
    )  
    crypto[col] = pd.to_numeric(crypto[col], errors="coerce")
```

```
# Handle missing values
```

```
for col in numeric_columns:
```

```
    crypto[col] = crypto[col].fillna(crypto[col].median())
```

```
# Create daily return
```

```
crypto["daily_return"] = (  
    (crypto["Close"] - crypto["Open"]) /  
    crypto["Open"]  
) * 100
```

```
# Create price range
```

```
crypto["price_range"] = crypto["High"] - crypto["Low"]
```

```
# Create volatility
```

```
crypto["volatility"] = (  
    (crypto["High"] - crypto["Low"]) /  
    crypto["Open"]  
) * 100
```

```
# Replace any remaining missing calculated values
```

```
crypto["daily_return"] = crypto["daily_return"].fillna(0)
```

```
crypto["price_range"] = crypto["price_range"].fillna(0)
```

```
crypto["volatility"] = crypto["volatility"].fillna(0)
```

```
# Standardize column names
```

```
crypto.columns = (  
    crypto.columns  
    .str.lower()  
    .str.strip()  
    .str.replace(" ", "_")  
)
```

```
print("Original Shape:", df.shape)
```

```
print("Cleaned Shape:", crypto.shape)
```

```
crypto.head()
```

```
print(crypto.isna().sum())

crypto.to_csv("crypto_cleaned.csv", index=False)

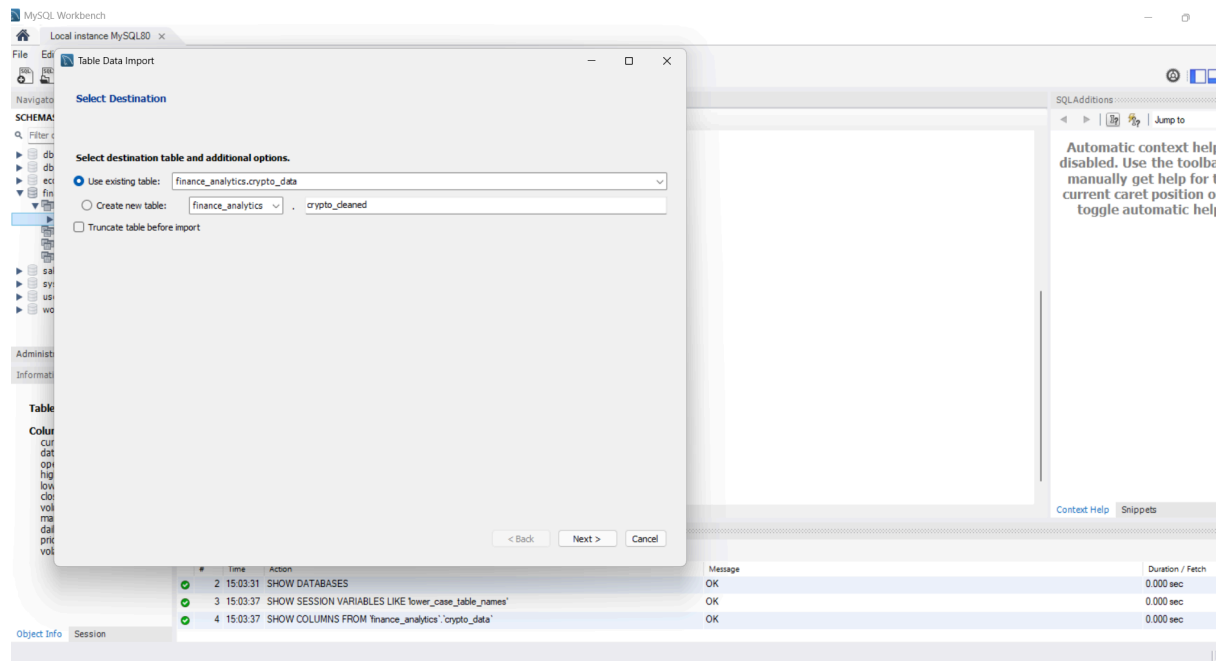
print("Cleaned dataset saved successfully.")
```

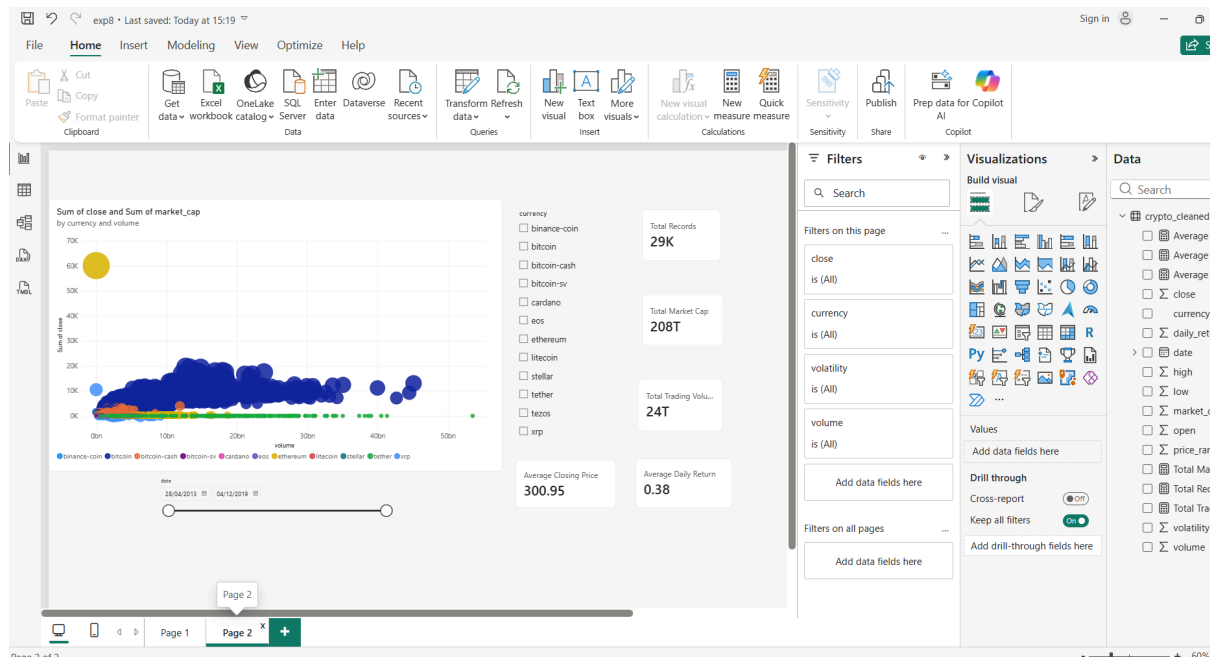
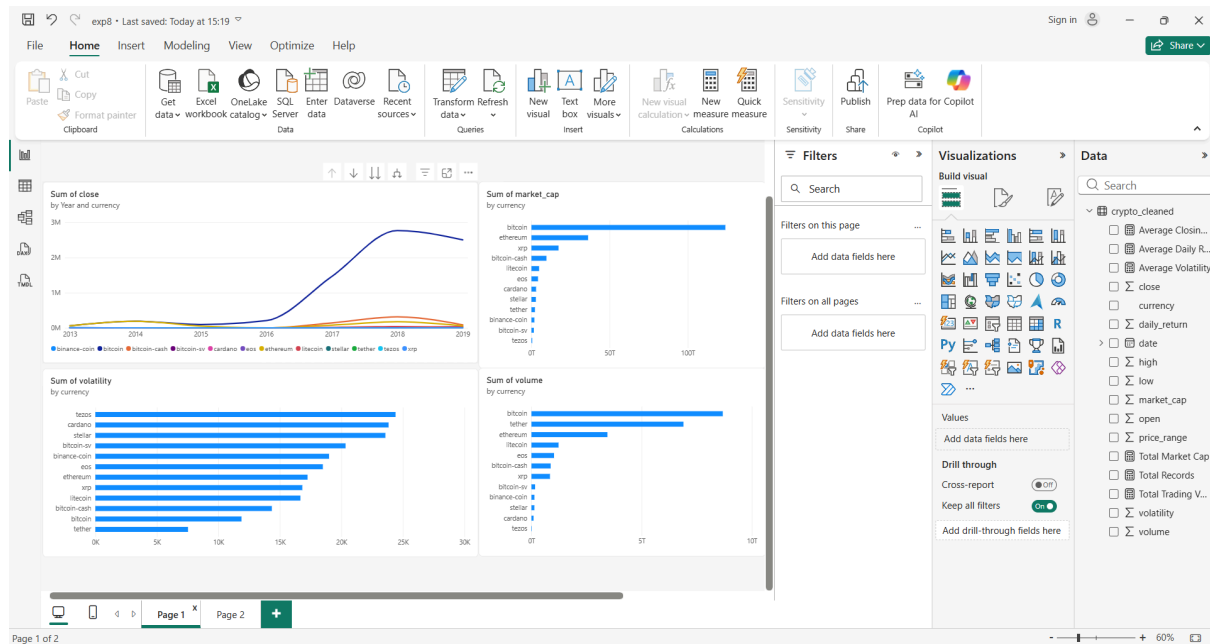
Output:

The screenshot shows a SQL query window with the following query: `FROM dbo.CryptoData_Clean;`. The results pane displays a table with 10 rows and 8 columns: Currency, Date, Open, High, Low, Close, Volume, and Market_Cap. The data represents daily price movements for Tezos (XZOS) from November 25, 2019, to December 4, 2019. A status bar at the bottom indicates 'Query completed with errors.' and 'localhost (17.0 RTM) | AAFRIN\91754 (72) | CryptoAnalytics | 00:00:00 | Row: 1, Col: 8'.

	Currency	Date	Open	High	Low	Close	Volume	Market_Cap
1	XZOS	Dec 04, 2019	1.29	1.32	1.25	1.25	46,048,752	824,588,509
2	XZOS	Dec 03, 2019	1.24	1.32	1.21	1.29	41,462,224	853,213,342
3	XZOS	Dec 02, 2019	1.25	1.26	1.20	1.24	27,674,097	817,672,179
4	XZOS	Dec 01, 2019	1.33	1.34	1.25	1.25	24,127,567	828,296,300
5	XZOS	Nov 30, 2019	1.31	1.37	1.31	1.33	28,706,667	879,181,680
6	XZOS	Nov 29, 2019	1.28	1.34	1.28	1.31	32,270,224	867,085,098
7	XZOS	Nov 28, 2019	1.26	1.35	1.22	1.28	44,240,281	845,073,679
8	XZOS	Nov 27, 2019	1.24	1.27	1.16	1.26	47,723,271	829,672,736
9	XZOS	Nov 26, 2019	1.24	1.28	1.23	1.24	54,828,808	822,065,277
10	XZOS	Nov 25, 2019	1.33	1.33	1.21	1.24	64,954,006	815,680,075

Total Rows: 10





Interpretation:

The dashboard provides a consolidated view of cryptocurrency market performance. The price trend visualization helps identify changes in cryptocurrency prices over time. Market capitalization comparisons show which cryptocurrencies have a larger market presence, while trading volume indicates the level of market activity. The volatility visualization helps identify cryptocurrencies with relatively higher price fluctuations and therefore higher market risk. The scatter plot allows the relationship between trading activity, closing price, and market capitalization to be observed.

Interactive filters allow users to analyze individual cryptocurrencies and selected time periods in greater detail.

Result:

The cryptocurrency dataset was successfully cleaned and transformed using Python and analyzed using Power BI. The dashboard effectively visualized cryptocurrency prices, market capitalization, trading volume, daily returns, and volatility. Interactive KPIs, slicers, filters, advanced visualizations, and drill-through functionality were implemented to support detailed financial analysis and provide meaningful insights into cryptocurrency market performance and risk.